



DEALER BEST PRACTICE SERIES

Hydraulic Stand For Disassembly of Rear Suspension Cylinders

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

Hydraulic Stand For Disassembly of Rear
Suspension Cylinders 0

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1.0 Introduction

The disassembly process for rear suspension cylinder takes one and a half hours approximately. In order to improve that time Ferreyros has implemented a hydraulic stand that allows them to do the same process in 20 minutes.

2.0 Best Practice Description

The focus of this Best Practice is improving the efficiency, safety, and ergonomics of suspension cylinder disassembly. The hydraulic stand can be used for two cylinders simultaneously (See picture 1). Using the stand, both cylinders are located vertically and the operator does not have to strain to handle the cylinders manually.



Picture 1. Suspension cylinder into hydraulic stand

Before this Best Practice, the cylinder was positioned horizontally (See the picture 2). That operation took a long time and the technician had to do more physical activity.

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Picture 2. Cylinder mounted before this best practice.

3.0 Implementation Steps

Disassemble Process using stand

- Use a sling in order to place the cylinder in the hydraulic stand.



Picture 3. Inside the fuel tank

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- Attach the cylinder with a pin in bottom level of the stand.



Picture 4. Cylinder cap clamped by a pin.



Picture 5. The top part of the cylinder is clamped by eight bolts and is adjustable for different sized cylinders

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- The bolts are removed using a pneumatic tool.



Picture 6. Bolt removed

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- Remove the Rod by pulling upward on the sling with a jib crane (See the picture 7)

**(1)****(2)****(3)****(4)**

Picture 7. Steps to remove the rod

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- Once the rod is removed, the inside diameter from the cylinder is measured with a micrometer (See picture 8) in order to check the dimensional tolerance. After that, a lifting bracket is bolted to the cylinder for removing the cylinder from the stand.

**(1)****(2)****(3)****(4)**

Picture 8. Steps to remove the cylinder

4.0 Benefits

- Safety operation – less manual handling.
- Reduce the disassembly time by 80%.
- Two suspension cylinders can be disassembled simultaneously.

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5.0 Resources Required

Fabrication shop.

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

None

8.0 Acknowledgements

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